6.9: Lab: Manage Security with SELinux

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3.1 Connexion à serverb avec les droits root

On se connecte en SSH à serverb depuis workstation avec le compte `student`, puis on passe root avec `sudo -i`. L'avertissement affiché signale seulement que serverb a été ajouté aux hôtes connus, puisqu'il s'agit de la première connexion.

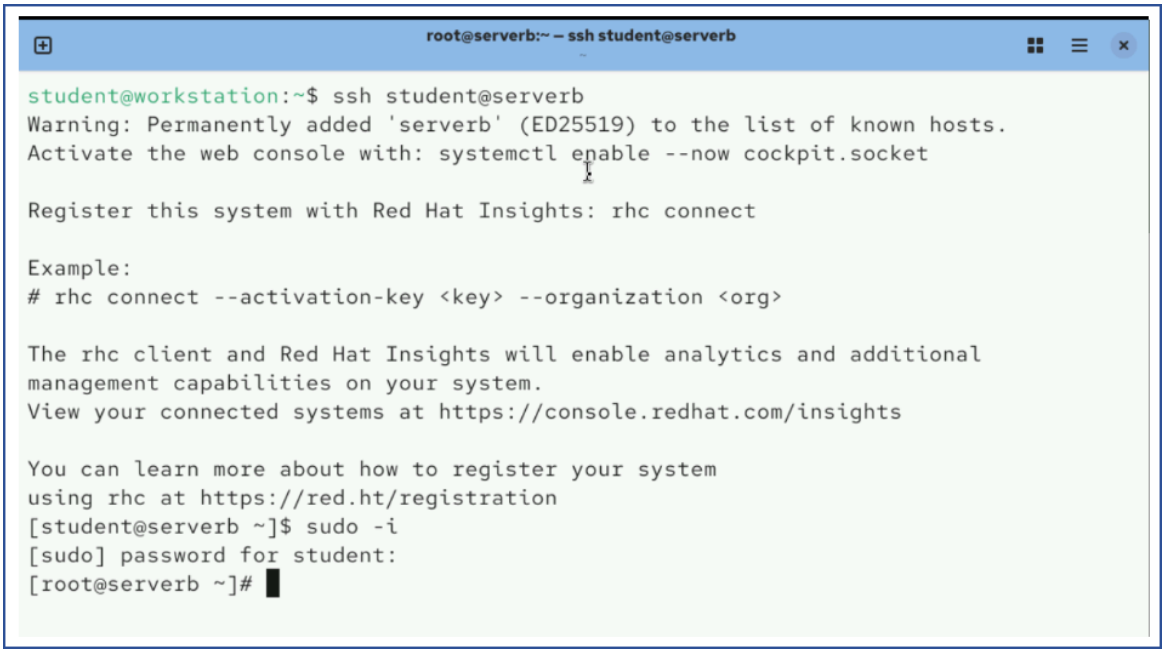


Figure 1 : session SSH vers serverb et sudo -i

3.2 Constat de l'erreur depuis workstation

L'ouverture de http://serverb/lab.html dans le navigateur renvoie une erreur **403 Forbidden** (« You don't have permission to access this resource »).



Figure 2 : erreur 403 Forbidden dans Firefox

3.3 Recherche de la cause dans les journaux

En filtrant `/var/log/messages` sur le mot « sealert », on trouve deux alertes de setroubleshoot : SELinux empêche `/usr/sbin/httpd` d'effectuer un accès getattr sur le fichier `/lab-content/lab.html`.

```
[root@serverb ~]# less /var/log/messages
[root@serverb ~]# less /var/log/messages | grep "sealert"
Oct 10 11:53:20 serverb setroubleshoot[2869]: SELinux is preventing /usr/sbin/httpd
 from getattr access on the file /lab-content/lab.html. For complete SELinux messag
es run: sealert -l 0e9df58d-e94b-4ee2-b162-f94a3ae0a81e
Oct 10 11:53:22 serverb setroubleshoot[2869]: SELinux is preventing /usr/sbin/httpd
 from getattr access on the file /lab-content/lab.html. For complete SELinux messag
es run: sealert -l 0e9df58d-e94b-4ee2-b162-f94a3ae0a81e
[root@serverb ~]#
```

Figure 3 : alertes setroubleshoot dans /var/log/messages

On analyse l'alerte avec `sealert -l 0e9df58d-e94b-4ee2-b162-f94a3ae0a81e`. Le plugin **catchall_labels** (confiance 83,8) recommande de modifier l'étiquette de `/lab-content/lab.html` à l'aide de `semanage fcontext` puis de `restorecon`.

```
[root@serverb ~]# sealert -l 0e9df58d-e94b-4ee2-b162-f94a3ae0a81e SELinux is preven
ting /usr/sbin/httpd from getattr access on the file lab-content/lab.html.
SELinux is preventing /usr/sbin/httpd from getattr access on the file /lab-content/
lab.html.

*****  Plugin catchall_labels (83.8 confidence) suggests   ********************

If you want to allow httpd to have getattr access on the lab.html file
Then you need to change the label on /lab-content/lab.html
Do
# semanage fcontext -a -t FILE_TYPE '/lab-content/lab.html'
where FILE_TYPE is one of the following: NetworkManager_dispatcher_chronyc_script_t
, NetworkManager_dispatcher_cloud_script_t, NetworkManager_dispatcher_console_scrip
t_t, NetworkManager_dispatcher_ddclient_script_t, NetworkManager_dispatcher_dhclien
t_script_t, NetworkManager_dispatcher_dnssec_script_t, NetworkManager_dispatcher_ex
ec_t, NetworkManager_dispatcher_iscsid_script_t, NetworkManager_dispatcher_script_t
, NetworkManager_dispatcher_sendmail_script_t, NetworkManager_dispatcher_tlp_script
_t, NetworkManager_dispatcher_winbind_script_t, NetworkManager_exec_t, NetworkManag
er_log_t, NetworkManager_priv_helper_exec_t, NetworkManager_tmp_t, abrt_dump_oops_e
xec_t, abrt_etc_t, abrt_exec_t, abrt_handle_event_exec_t, abrt_helper_exec_t, abrt_
retrace_coredump_exec_t, abrt_retrace_spool_t, abrt_retrace_worker_exec_t, abrt_tmp
_t, abrt_upload_watch_tmp_t, abrt_var_cache_t, abrt_var_log_t, abrt_var_run_t, acco
untsd_exec_t, acct_data_t, acct_exec_t, admin_crontab_tmp_t, admin_passwd_exec_t, a
fs_logfile_t, aide_exec_t, aide_log_t, alsa_exec_t, alsa_tmp_t, amanda_exec_t, aman
```

Figure 4 : sealert : proposition du plugin catchall_labels

Le plugin **catchall** (confiance 17,1) propose uniquement de créer un module de politique local ; cette solution est écartée. Les informations complémentaires indiquent un contexte source `httpd_t`, un contexte cible `default_t` et le mode Enforcing.

```
*****  Plugin catchall (17.1 confidence) suggests   **************************

If you believe that httpd should be allowed getattr access on the lab.html file by
default.
Then you should report this as a bug.
You can generate a local policy module to allow this access.
Do
allow this access for now by executing:
# ausearch -c 'httpd' --raw | audit2allow -M my-httpd
# semodule -X 300 -i my-httpd.pp

Additional Information:
Source Context                system_u:system_r:httpd_t:s0
Target Context                unconfined_u:object_r:default_t:s0
Target Objects                /lab-content/lab.html [ file ]
Source                        httpd
Source Path                   /usr/sbin/httpd
Port                          <Unknown>
Host                          serverb
Source RPM Packages           httpd-core-2.4.63-1.el10.x86_64
Target RPM Packages
SELinux Policy RPM            selinux-policy-targeted-40.13.26-1.el10.noarch
Local Policy RPM              selinux-policy-targeted-40.13.26-1.el10.noarch
```

Figure 5 : plugin catchall et informations complémentaires

```
Local Policy RPM              selinux-policy-targeted-40.13.26-1.el10.noarch
Selinux Enabled               True
Policy Type                   targeted
Enforcing Mode                Enforcing
Host Name                     serverb
Platform                      Linux serverb 6.12.0-55.12.1.el10_0.x86_64 #1 SMP
                              PREEMPT_DYNAMIC Tue May 13 02:51:44 EDT 2025
                              x86_64
Alert Count                   2
First Seen                    2026-10-10 11:53:18 UTC
Last Seen                     2026-10-10 11:53:18 UTC
Local ID                      0e9df58d-e94b-4ee2-b162-f94a3ae0a81e

Raw Audit Messages
type=AVC msg=audit(1791633198.32:848): avc:  denied  { getattr } for  pid=2603 comm
="httpd" path="/lab-content/lab.html" dev="sda3" ino=16998947 scontext=system_u:sys
tem_r:httpd_t:s0 tcontext=unconfined_u:object_r:default_t:s0 tclass=file permissive
=0


type=SYSCALL msg=audit(1791633198.32:848): arch=x86_64 syscall=newfstatat success=n
o exit=EACCES a0=ffffff9c a1=7f5078004a98 a2=7f5087ffe9f0 a3=100 items=0 ppid=2599
pid=2603 auid=4294967295 uid=48 gid=48 euid=48 suid=48 fsuid=48 egid=48 sgid=48 fsg
id=48 tty=(none) ses=4294967295 comm=httpd exe=/usr/sbin/httpd subj=system_u:system
_r:httpd_t:s0 key=(null)

Hash: httpd,httpd_t,default_t,file,getattr

[root@serverb ~]#
```

Figure 6 : messages d'audit bruts : AVC denied getattr (2 alertes)

La commande `ausearch -m AVC -ts recent` confirme les deux refus : processus httpd, fichier `/lab-content/lab.html`, accès de `httpd_t` vers `default_t`.

```
[root@serverb ~]# ausearch -m AVC -ts recent
----
time->Sat Oct 10 11:53:18 2026
type=PROCTITLE msg=audit(1791633198.031:847): proctitle=2F7573722F7362696E2F6874747
064002D44464F524547524F554E44
type=SYSCALL msg=audit(1791633198.031:847): arch=c000003e syscall=262 success=no ex
it=-13 a0=ffffff9c a1=7f50780049c0 a2=7f5087ffe9f0 a3=0 items=0 ppid=2599 pid=2603
auid=4294967295 uid=48 gid=48 euid=48 suid=48 fsuid=48 egid=48 sgid=48 fsgid=48 tty
=(none) ses=4294967295 comm="httpd" exe="/usr/sbin/httpd" subj=system_u:system_r:ht
tpd_t:s0 key=(null)
type=AVC msg=audit(1791633198.031:847): avc:  denied  { getattr } for  pid=2603 com
m="httpd" path="/lab-content/lab.html" dev="sda3" ino=16998947 scontext=system_u:sy
stem_r:httpd_t:s0 tcontext=unconfined_u:object_r:default_t:s0 tclass=file permissiv
e=0
----
time->Sat Oct 10 11:53:18 2026
type=PROCTITLE msg=audit(1791633198.032:848): proctitle=2F7573722F7362696E2F6874747
064002D44464F524547524F554E44
type=SYSCALL msg=audit(1791633198.032:848): arch=c000003e syscall=262 success=no ex
it=-13 a0=ffffff9c a1=7f5078004a98 a2=7f5087ffe9f0 a3=100 items=0 ppid=2599 pid=260
3 auid=4294967295 uid=48 gid=48 euid=48 suid=48 fsuid=48 egid=48 sgid=48 fsgid=48 t
ty=(none) ses=4294967295 comm="httpd" exe="/usr/sbin/httpd" subj=system_u:system_r:
httpd_t:s0 key=(null)
type=AVC msg=audit(1791633198.032:848): avc:  denied  { getattr } for  pid=2603 com
m="httpd" path="/lab-content/lab.html" dev="sda3" ino=16998947 scontext=system_u:sy
stem_r:httpd_t:s0 tcontext=unconfined_u:object_r:default_t:s0 tclass=file permissiv
e=0
[root@serverb ~]#
```

Figure 7 : refus AVC récents avec ausearch

3.4 Correction du contexte

La comparaison `ls -dZ /lab-content /var/www/html` montre que le nouveau répertoire est de type `default_t`, alors que le répertoire web d'origine est de type `httpd_sys_content_t`. On ajoute donc la règle `semanage fcontext -a -t httpd_sys_content_t '/lab-content(/.*)?'`, puis on l'applique avec `restorecon -Rv /lab-content/` : le répertoire et le fichier `lab.html` sont réétiquetés.

```
[root@serverb ~]# ls -dZ /lab-content /var/www/html
      unconfined_u:object_r:default_t:s0 /lab-content
system_u:object_r:httpd_sys_content_t:s0 /var/www/html
[root@serverb ~]# semanage fcontext -a -t httpd_sys_content_t '/lab-content(/.*)?'
[root@serverb ~]# restorecon -Rv /lab-content/
Relabeled /lab-content from unconfined_u:object_r:default_t:s0 to unconfined_u:obje
ct_r:httpd_sys_content_t:s0
Relabeled /lab-content/lab.html from unconfined_u:object_r:default_t:s0 to unconfin
ed_u:object_r:httpd_sys_content_t:s0
[root@serverb ~]#
```

Figure 8 : comparaison des contextes, semanage fcontext et restorecon

3.5 Vérification depuis workstation

Après rechargement, la page http://serverb/lab.html affiche « This is the html file for the SELinux final lab on SERVERB. ».

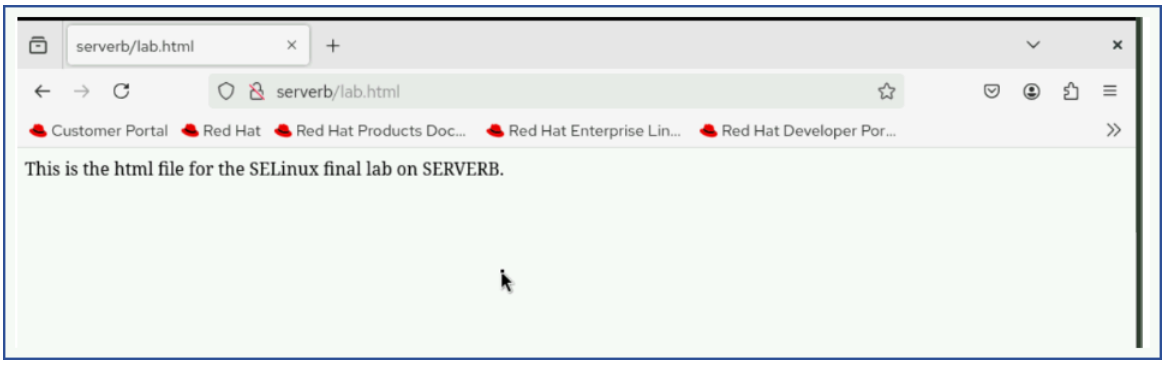


Figure 9 : la page lab.html s'affiche correctement

3.6 Évaluation

Sur workstation, `lab grade selinux-review` renvoie **PASS** pour la vérification de l'accès à http://serverb/lab.html.

```
student@workstation:~$ lab grade selinux-review
🤖 Running: lab grade selinux-review
PASS    Check access to http://serverb/lab.html
student@workstation:~$
```

Figure 10 : résultat de lab grade selinux-review